

# Two genuine rotors for one preserving basis change

## Conjecture 00000007629

earthking11

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### Statement and precise scope

The conjecture states that rotor realizations of orthogonal basis changes are unique, with orientation reversal as an exception. We disprove literal uniqueness using a change that preserves orientation. No identification of opposite rotors appears in the source; a uniqueness statement in the quotient by the central sign is a different assertion and is not refuted here.

**Theorem 1.** *For the Euclidean plane  $V = \mathbb{R}^2$ ,  $Q(v) = v_0^2 + v_1^2$ , the actual spin group in its Clifford algebra contains two distinct elements 1 and  $-1$ , both realizing the identity orthonormal-basis change. The change has determinant 1, so orientation reversal does not account for this failure of uniqueness.*

*Proof.* Let  $A = \text{Cl}(V, Q)$  and write  $\iota : V \rightarrow A$  for the vector inclusion. The Clifford relation is  $\iota(v)^2 = Q(v)1$ . The standard basis vectors  $e_0, e_1$  have  $Q(e_i) = 1$  and zero polar pairing when their indices differ. Thus this is an actual orthonormal basis for the specified form.

Both  $\iota(e_0)$  and  $\iota(-e_0)$  are units, each its own inverse. They belong to the vector-unit generators of the Lipschitz group, and

$$\iota(e_0)\iota(-e_0) = -\iota(e_0)^2 = -1.$$

Thus  $-1$  belongs to the Lipschitz group and is even. Its Clifford conjugate is itself, so its product with that conjugate is 1. This proves pin unitarity, and hence membership in the actual spin group. The unit 1 also belongs to this group. The real scalar map into  $A$  is injective; since  $2 \neq 0$  in  $\mathbb{R}$ , these two elements are distinct.

A rotor  $r$  realizes a map  $T$  here precisely when  $r\iota(v)\text{star}(r) = \iota(Tv)$  for every vector  $v$ . Direct calculation gives

$$1\iota(v)\text{star}(1) = \iota(v), \quad (-1)\iota(v)\text{star}(-1) = \iota(v).$$

Both rotors therefore realize  $T = \text{id}_V$ . This map fixes every standard basis vector, preserves  $Q$ , and has determinant 1. Since the two realizations are distinct, there cannot be a unique realizing rotor.  $\square$

### Formal verification

The Lean project uses Mathlib's actual `CliffordAlgebra` and `spinGroup`, not a substitute group. It proves the membership of  $-1$  through generating vector units, pin unitarity and the even subalgebra. It verifies the squared-norm formula, basis orthonormality, all-vector conjugation identities, and determinant. The capstone includes these facts and  $\neg\exists!r \in \text{Spin}(Q)$  realizing the identity.

The independently checked project builds with Lean 4.33.1 and Mathlib at the fixed commit recorded in the manifest. Its printed axiom audits use only propositional extensionality, classical choice and quotient soundness; there are no unfinished proofs, native evaluation or additional axioms. No numerical search or long brute-force computation is used.